Ingeniería de soluciones con AI

Semestre 2026-1

Docente:

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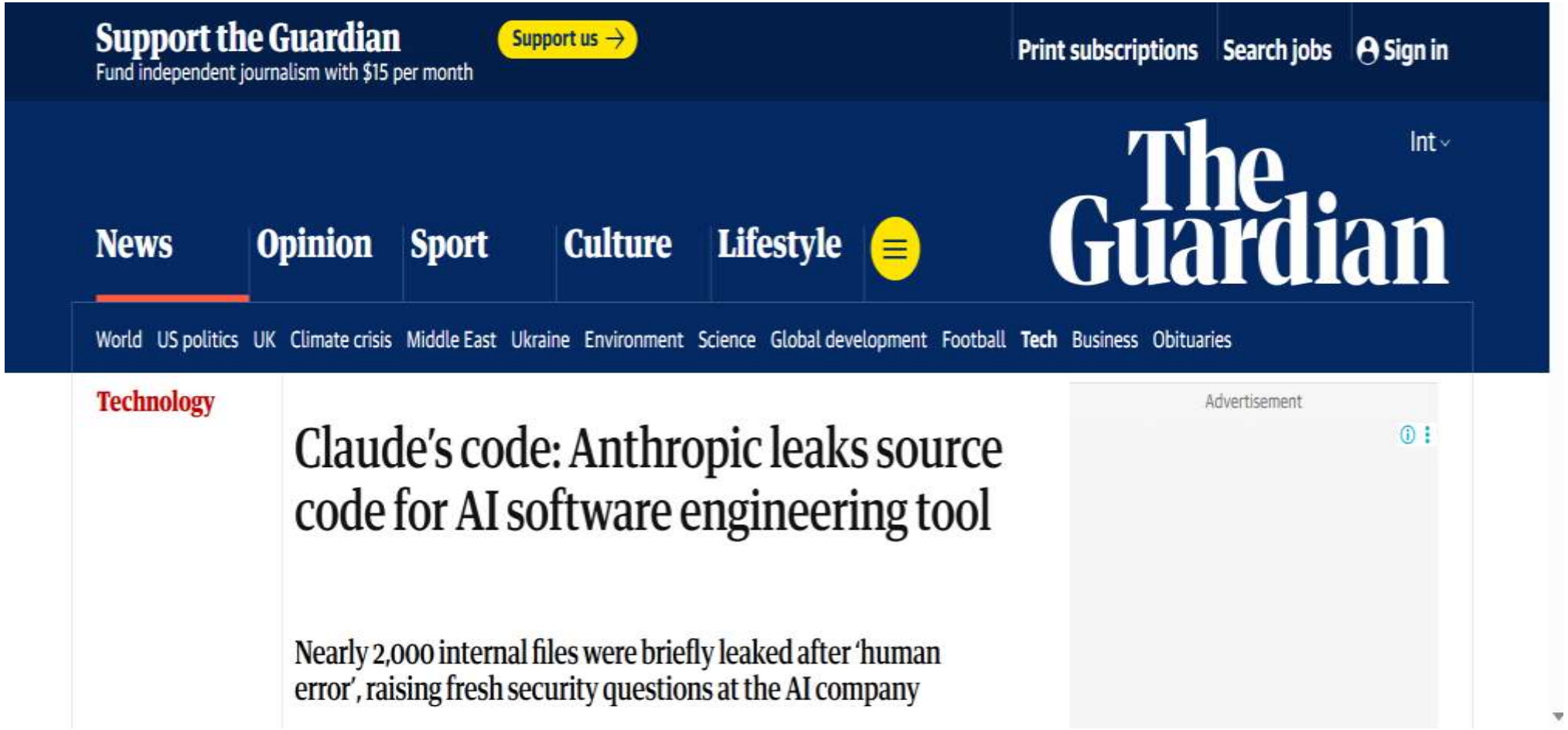
***Noticias**: Some files were leaked in Claude Code.*



¿Qué es un agente?

“An agent is anything that can perceive its environment and act upon that environment.”

Nota: *Huyen, C. (2025). AI engineering. O'Reilly Media*

¿Cuáles son sus componentes?

Memoria

mecanismos que permiten a un modelo retener y utilizar información.

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Plan

Capacidad del modelo de descomponer un problema en tareas para resolverlo.

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LLMs

El motor del modelo. Los principales proveedor son OpenAI, Google y Anthropic

Tools

Funciones que le permiten al modelo conectarse con el mundo exterior.

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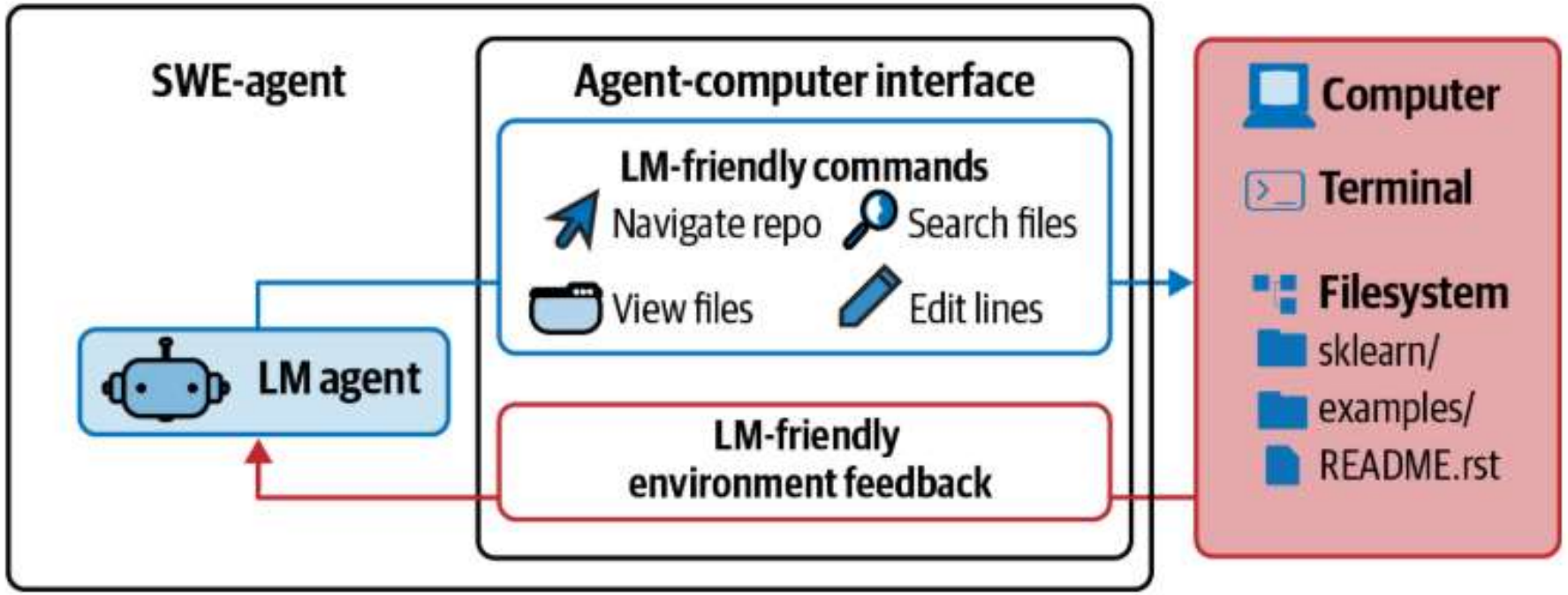


Figure 6-8. SWE-agent (Yang et al., 2024) is a coding agent whose environment is the computer and whose actions include navigation, search, and editing. Adapted from an original image licensed under CC BY 4.0.

Los agentes de código tienen herramientas que les permiten conectarse a los archivos (comandos de Linux).

Tarea:
Proyección de las ventas para Fruity Fedora sobre los próximos tres años.

1. Reason about how to accomplish this task. It might decide that to predict future sales, it first needs the sales numbers from the last five years. Note that the agent's reasoning is shown as its intermediate response.
2. Invoke SQL query generation to generate the query to get sales numbers from the last five years.
3. Invoke SQL query execution to execute this query.
4. Reason about the tool outputs and how they help with sales prediction. It might decide that these numbers are insufficient to make a reliable projection, perhaps because of missing values. It then decides that it also needs information about past marketing campaigns.
5. Invoke SQL query generation to generate the queries for past marketing campaigns.
6. Invoke SQL query execution.
7. Reason that this new information is sufficient to help predict future sales. It then generates a projection.
8. Reason that the task has been successfully completed.

Nota: *Huyen, C. (2025). AI engineering. O'Reilly Media*

LLMs: *Clave de los agentes.*

Model	Features: Context Window	Creator	Intelligence: Artificial Analysis Intelligence Index	Price: Blended USD/1M Tokens	Speed: Median Tokens/s	Latency: Latency First Chunk (s)	End-to-End Response Time: Total Response (s)	Further Analysis
Gemini 3.1 Pro Preview	1M	Google	57	$4.50	136	32.51	36.20	Model Providers
GPT-5.4 (xhigh)	1.05M	OpenAI	57	$5.63	74	163.52	170.27	Model Providers
GPT-5.3 Codex (xhigh)	400k	OpenAI	54	$4.81	75	62.24	68.91	Model Providers
Claude Opus 4.6 (max)	1M	Anthropic	53	$10.00	43	9.22	20.89	Model Providers
Claude Sonnet 4.6 (max)	1M	Anthropic	52	$6.00	72	107.86	114.83	Model Providers
GPT-5.2 (xhigh)	400k	OpenAI	51	$4.81	76	105.36	111.91	Model Providers
GLM-5	200k	Z AI	50	$1.55	78	1.66	47.70	Model Providers
Claude Opus 4.5	200k	Anthropic	50	$10.00	47	11.16	21.80	Model Providers
MiniMax-M2.7	205k	MiniMax	50	$0.53	53	2.10	58.52	Model Providers
MiMo-V2-Pro	1M	Xiaomi	49	$1.50	--	--	--	Model Providers

Nota: *https://artificialanalysis.ai/leaderboards/models*

Plan: *Divida las tareas complejas en subtareas más simples.*

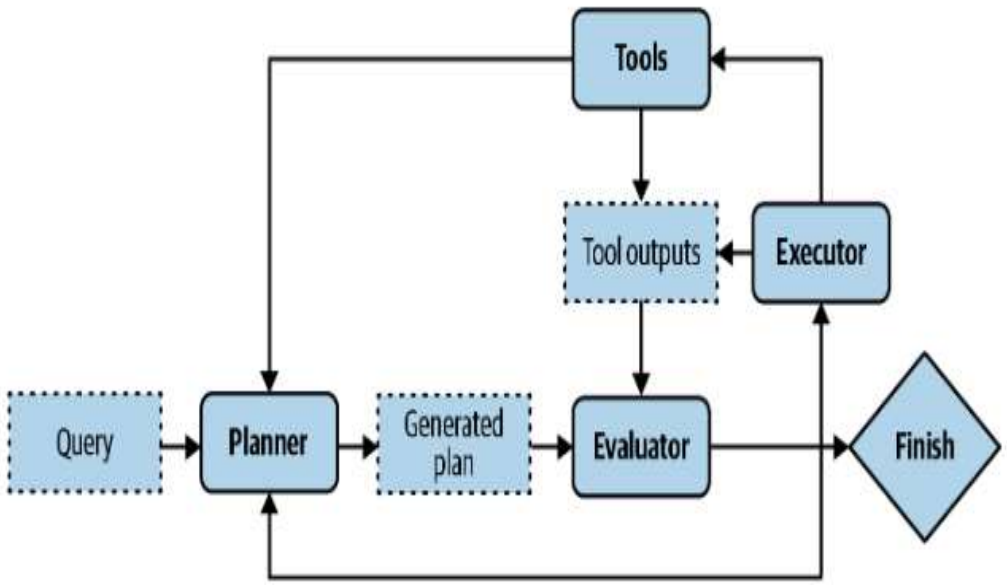


Figure 6-9. Decoupling planning and execution so that only validated plans are executed.

1. *Plan generation*: come up with a plan for accomplishing this task. A plan is a sequence of manageable actions, so this process is also called task decomposition.
2. *Reflection and error correction*: evaluate the generated plan. If it's a bad plan, generate a new one.
3. *Execution*: take the actions outlined in the generated plan. This often involves calling specific functions.
4. *Reflection and error correction*: upon receiving the action outcomes, evaluate these outcomes and determine whether the goal has been accomplished. Identify and correct mistakes. If the goal is not completed, generate a new plan.

Nota: *https://artificialanalysis.ai/leaderboards/models*

Tools: *La conexión con el entorno.*

```
def lbs_to_kg(lbs):
    return lbs * 0.45359237

def ft_to_meters(ft):
    return ft * 0.3048
```

Tool definition

```
lbs_to_kg_tool = FunctionDeclaration(
    name="lbs_to_kg",
    description="Convert from pounds to kilograms",
    parameters={
        "type": "object",
        "properties": {
            "lbs": {"type": "int", "description": "The value to be converted"}
        },
        "required": ["lbs"]
    },
)

ft_to_m_tool = FunctionDeclaration(name="ft_to_meters",...)
```

(1) Tool description

```
messages = [{"role": "user", "content": [USER_QUERY]}]
```

User query

```
response = model_client.chat.completions.create(
    model=[MODEL_NAME],
    messages=messages,
    tools=[lbs_to_kg_tool, ft_to_m_tool],
    tool_choice="auto",
)
```

(2) This query can use two tools

Figure 6-10. An example of a model using two simple tools.

Function calling es una forma de llamar tools usando modelos de lenguaje.

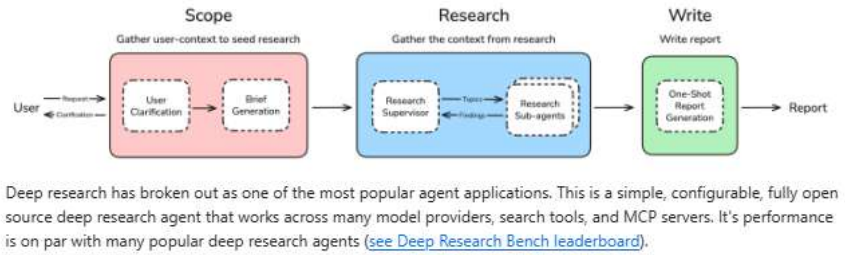
Los principales consejos que hay que tener al momento de usar tools son:

1- Definir correctamente el nombre de la tool.
2- Definir la funcionalidad de la tool (descripción).
3- Definir correctamente el nombre de los argumentos.
4- Definir el tipo de dato de los argumentos.
5- Definir los parámetros requeridos.

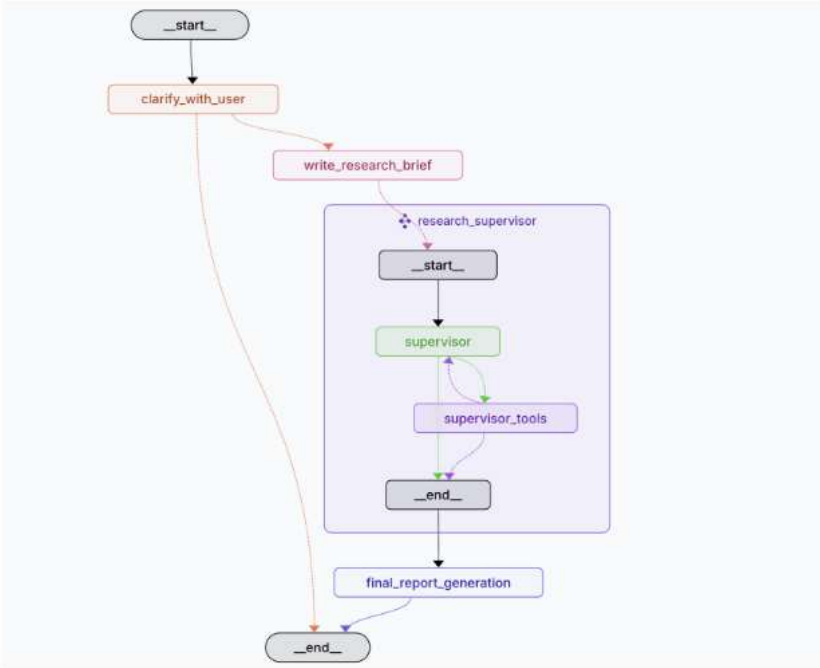
Los modelos usan, seleccionan e implementan mejor las tools con mayor información.

Nota: *Huyen, C. (2025). AI engineering. O'Reilly Media*

Agente : Búsqueda profunda.



Deep research has broken out as one of the most popular agent applications. This is a simple, configurable, fully open source deep research agent that works across many model providers, search tools, and MCP servers. It's performance is on par with many popular deep research agents (see Deep Research Bench leaderboard).



Arquitectura agentica propuesta por LangChain para hacer una búsqueda profunda.

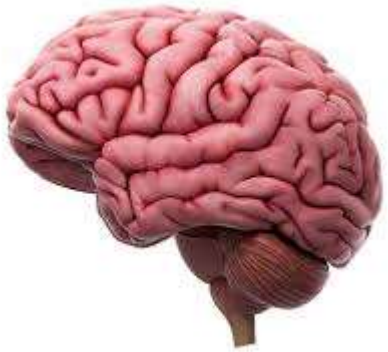
Esta arquitectura tiene tres etapas:

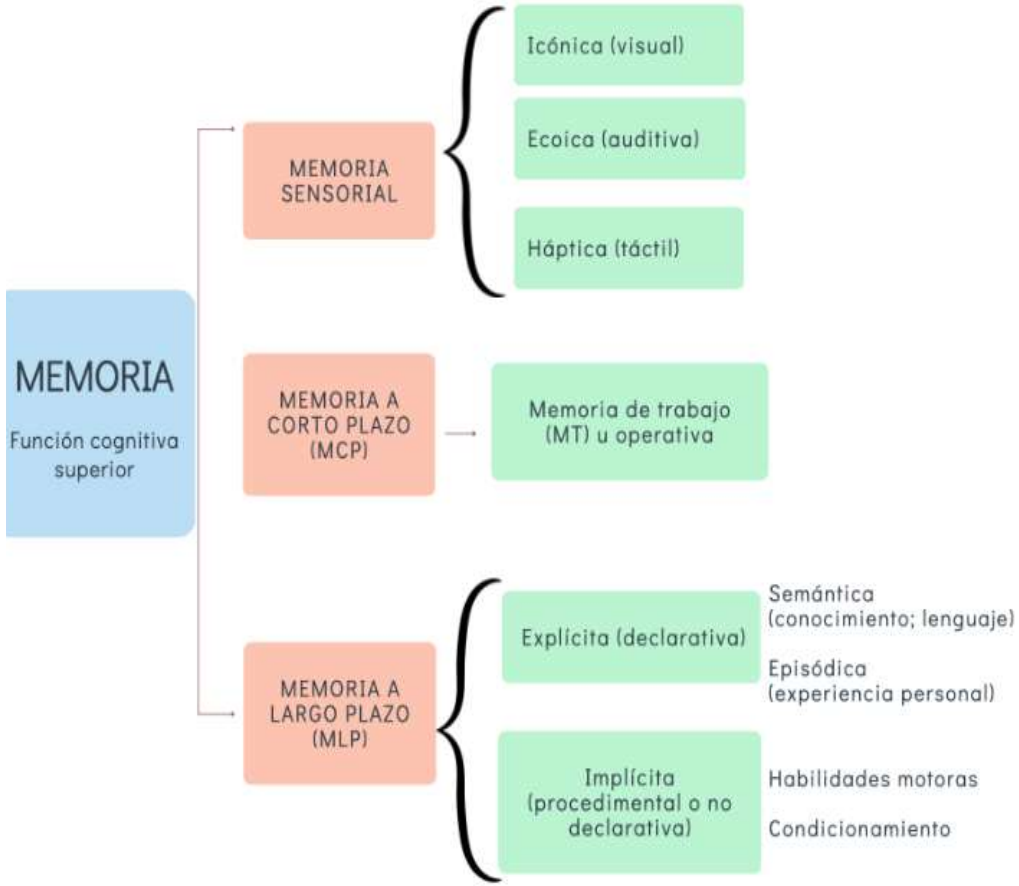
1. Scope
2. Research
3. Write

Las etapas están compuestas de nodos, que tienen tools y capacidades de razonamiento.

Nota:*https://github.com/langchain-ai/open_deep_research*

Memoria : *Próxima clase…*





Si fueran agentes, ¿Qué memoria seria útil tener?

Agentes *más que solo LLMs.*

Memoria

mecanismos que permiten a un modelo retener y utilizar información.

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Plan

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